

Fourier Domain Detection of AI Generated Images

1 Image Representation

Let the grayscale image be represented as:

$$I(x, y)$$

where

$$x = 0, 1, 2, \dots, W - 1$$

$$y = 0, 1, 2, \dots, H - 1$$

and

$$I(x, y) \in [0, 255]$$

2 2D Discrete Fourier Transform

The 2D Discrete Fourier Transform (DFT) of the image is:

$$F(u, v) = \sum_{x=0}^{W-1} \sum_{y=0}^{H-1} I(x, y) e^{-j2\pi(\frac{ux}{W} + \frac{vy}{H})}$$

where

$$u = 0, 1, 2, \dots, W - 1$$

$$v = 0, 1, 2, \dots, H - 1$$

3 FFT Shift

To center the DC component:

$$F_{shift}(u, v) = F((u + W/2) \bmod W, (v + H/2) \bmod H)$$

This moves the zero-frequency component to the center.

4 Magnitude Spectrum

Magnitude is computed as:

$$M(u, v) = |F_{shift}(u, v)|$$

Log magnitude is used for visualization:

$$M_{log}(u, v) = 20 \log (1 + |F_{shift}(u, v)|)$$

5 Phase Spectrum

The phase spectrum is:

$$\phi(u, v) = \angle F_{shift}(u, v)$$

$$\phi(u, v) = \tan^{-1} \left(\frac{\text{Im}(F_{shift}(u, v))}{\text{Re}(F_{shift}(u, v))} \right)$$

6 Power Spectrum

The power spectrum is:

$$P(u, v) = |F_{shift}(u, v)|^2$$

7 Radial Distance from DC

Let the center be:

$$c_x = \frac{W}{2}$$

$$c_y = \frac{H}{2}$$

Radial distance:

$$R(u, v) = \sqrt{(u - c_x)^2 + (v - c_y)^2}$$

8 Radial Power Spectrum

Radial mean power:

$$P_r(r) = \frac{1}{N_r} \sum_{R(u,v)=r} P(u, v)$$

Where:

N_r = number of pixels with radius r

9 DC Component Energy

DC component:

$$P_{DC} = P(c_x, c_y)$$

Total energy:

$$P_{total} = \sum_{u,v} P(u, v)$$

DC energy fraction:

$$DC_{fraction} = \frac{P_{DC}}{P_{total}} \times 100$$

10 Grid Artifact Detection

Suppress center region:

$$M_{suppressed}(u, v) = 0 \quad \text{if} \quad R(u, v) < R_s$$

Otherwise:

$$M_{suppressed}(u, v) = M(u, v)$$

Threshold:

$$T = \text{Percentile}(M_{suppressed}, \alpha)$$

Peak detection:

$$Peak(u, v) = \begin{cases} 1 & M_{suppressed}(u, v) > T \\ 0 & \text{otherwise} \end{cases}$$

Peak count:

$$N_{peaks} = \sum_{u,v} Peak(u, v)$$

11 Spectrum Flatness

Radial mean:

$$\mu = \text{mean}(P_r)$$

Radial standard deviation:

$$\sigma = \text{std}(P_r)$$

Flatness metric:

$$Flatness = 1 - \frac{\sigma}{\mu + \epsilon}$$

Where

$$\epsilon \approx 10^{-8}$$

12 AI Detection Score

Define binary indicators:

$$S_1 = \begin{cases} 1 & DC_{fraction} < 5 \\ 0 & \text{otherwise} \end{cases}$$

$$S_2 = \begin{cases} 1 & N_{peaks} > 0.01 \times (W \times H) \\ 0 & \text{otherwise} \end{cases}$$

$$S_3 = \begin{cases} 1 & Flatness > 0 \\ 0 & \text{otherwise} \end{cases}$$

Final AI score:

$$Score = S_1 + S_2 + S_3$$

13 Final Classification

$$Score = 0 \rightarrow \text{Natural Image}$$

$$Score = 1 \rightarrow \text{Ambiguous}$$

$$Score = 2 \rightarrow \text{Probable AI}$$

$$Score = 3 \rightarrow \text{Strong AI}$$

14 Summary

$I(x, y) \rightarrow FFT \rightarrow Magnitude \rightarrow PowerSpectrum \rightarrow RadialAnalysis \rightarrow PeakDetection \rightarrow AIScore$